PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

REMATERIAL POST

返料岗位

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the rematerial post under complete closure.

此程序涵盖操作员对完全关闭下的返料岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the rematerial post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the rematerial post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对返料岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定返料岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP岗位操作规程
- Belt Conveyors(4303-BCV-002). 胶带机(4303-BCV-002)
- Bag Filter (4301-DUC-201). 布袋除尘器 (4301-DUC-201)
- Fan (4301-FAN-201). 风机 (4301-FAN-201)

4. PPE 个人保护装备



耳塞　安全帽　眼镜　劳保鞋　手套　工作服



防尘口罩

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5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Dust; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Object falling from high altitudes; 7. Noise. 1. 机械伤害; 2. 灰尘; 3. 触电; 4. 湿滑的条件，绊倒危险; 5. 窒息; 6. 高空坠物; 7. 噪音。	1. Install a protective cover on the rotating part of the equipment; 2. Wear dust mask as required; 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clear clearance or maintenance operations; 6. Wear labor protection goods and check the safety of stacking items on the operation platform; 7. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩; 2. 按要求佩戴防尘口罩; 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效; 4. 上下楼梯请扶好扶手; 5. 清仓或维修作业执行审批程序; 6. 穿戴好劳保护品，检查各平台的物品堆放安全性; 7. 佩戴好耳塞，保护听力。

6. Equipment 工具/ 设备

	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训;

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№	Stage	Key Points
		2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患； 3. 检查并打扫设备、现场卫生； 5. 4.与前后工序取得联系，并发出明确指令。
2	Inspect Belt Conveyors(4303-BCV-002) 检查胶带机(4303-BCV-002)	2. Each bearing is well lubrication and closed, the reducer oil level is appropriate, the upper and lower roller lubrication is good, the operation is flexible, whether the wear of the first and tail wheels and the tape are filled with materials, and whether the cleaning device of the first and tail wheels is in good condition; 3. Comprehensively check whether all parts of the belt conveyors are intact, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 4. Whether the electrical facilities and control system are intact, whether the protective device is firm, and whether the funnel baffle and liner are damaged; 5. Check whether the cushion rollers of the impact load parts are intact; 6. Check the installation of belt sweeper; 7. Whether the installed electrical facilities and signal devices are intact and responsive; 8. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, and watch whether the running direction of the tape is correct, and whether the belt can run away; 9. After checking and contacting the post, press the start button; 10. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 1. 各部轴承润滑与封闭良好，减速机油位适当，上下托滚润滑良好，运转灵活，首、尾轮磨损情况与胶带间是否夹有物料，首、尾轮的清扫装置是否完好； 2. 全面检查胶带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好； 3. 电器设施和控制系统是否完好，防护装置是否牢固，漏斗挡板和衬板是否损坏； 4. 检查承受冲击载荷部位的缓冲托辊是否完好； 5. 检查胶带机清扫器安装是否正常；

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№	Stage	Key Points
		6. 装设的电器设施及信号装置是否完好和反应灵敏； 7. 将现场控制盒置于手动位置，按一下启动按钮，迟缓旋动电位器，观看胶带运转方向能否正确，皮带是否跑偏； 8. 检查无误并与前后岗位联系后，按下启动按钮； 9. 运转过程中，检查胶带机急停装置、皮带纠偏和测速传感器是否控制灵敏。
3	Inspect Bag Filter (4301-DUC-201) 检查布袋除尘器 (4301-DUC-201)	1. Whether the electrical high and low voltage system and PLC control are normal; 2. Whether the inlet valve of the filter chamber of each unit is open; 3. Whether all the solenoid pulse valves are normal and the action sequence is correct; 4. All measurement circuits are normal and signal; 5. All control valves are in the automatic mode and in the desired position; 6. Open all inspection entrances and doors to check whether the equipment moving parts are stuck in dust or external objects; after inspection, close all inspection population and doors and check all pipes and equipment inspection holes for closure; 7. Whether the sealing strip of the air chamber is sealed, and whether the fastening bolt of air box sealing cover is in good condition; 8. Connect the pressure measuring device to the pressure measuring port of the bag filter; 9. Check whether the bag back blowing pressure is 0.4MPa~0.5MPa; 10. Check whether the water drain trap is normal; 11. Check the operation condition of the ash removal device of the bag filter. 1. 电气高低压系统、PLC控制是否正常； 2. 除尘器各单元过滤室的进口阀门是否都打开； 3. 所有电磁脉冲阀是否正常，动作顺序正确； 4. 所有测量回路是否正常、有信号； 5. 所有提升阀处于自动模式和需要的位置； 6. 打开所有检查入口、门，检查设备运动部件是否被灰尘或外来物卡住；检查后，关闭所有检查人口、门，并检查所有管道和设备检查孔是否关闭； 7. 反吹风气室密封条是否是密封完好，气箱密封盖紧固螺栓是否完好； 8. 接通测压装置到布袋除尘器的测压口；

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№	Stage	Key Points
		9. 检查布袋反吹风压力是否是0.4MPa ~ 0.5MPa； 10. 检查疏水阀是否正常； 11. 检查布袋除尘器排灰装置工作情况。
4	Inspect Fan (4301-FAN-201) 检查风机 (4301-FAN-201)	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena; 2. The fan air outlet valve is normal, and the fan inlet valve is closed; 3. Whether the coupling concentricity meets the requirements; 4. Check whether the lubricating oil level of the fan bearing box is normal; 5. Close the power switch, first click (press the operation button and quickly press the stop button) to confirm that the steering is correct, no noise, no vibration and other abnormal conditions; 6. After the fan is started and running, observe the change of the fan sound, check the temperature and noise of the fan bearing and the motor bearing, and check the fan vibration (including the impeller). 1. 手动盘车2 ~ 3转，转动灵活，不得有刮蹭、卡住等异常现象； 2. 风机出风阀门正常，风机进口阀门关闭； 3. 联轴器同心度是否符合要求； 4. 检查风机轴承箱润滑油位是否正常； 5. 合上电源开关，先点动(按运转按钮后快速按停止按钮)确认转向正确、无杂音、无振动等异常情况； 6. 风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的温度和噪声，检查风机振动 (包括叶轮)。
5	Check the other Equipment and Facilities 检查其它设备设施	1. Import and outlet valve of fan (4301-FAN-201) a. Check whether the inlet and outlet valve (****) is normal. 1. 风机 (4301-FAN-201)进出口阀门 a. 检查进出阀门(****)是否正常。 2. Belt machine (4303-BCV-002), bag filter (4301-DUC-201) a. Check whether the feeding chute is in good condition; b. Check whether the blocking material detection switch of the lower feeding chute is normal. 2. 胶带机(4303-BCV-002)下料溜槽，布袋除尘器 (4301-

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№	Stage	Key Points
		DUC-201)下料溜槽 a. 检查各下料溜槽是否完好； b. 检查下料溜槽的堵料检测开关是否正常。 3. Star Discharger a. Check the correct rotation direction of the star valve rotor under the bag filter (4301-DUC-201). 3. 耐温卸料器 a. 检查布袋除尘器 (4301-DUC-201)下方耐温卸料器转子的旋转方向是否正确。
6	Inspect Materials Level 检查物料料位	1. Check for material storage on the tape machine (4303-BCV-002), and no adhesive deposit on the surface of the head and tail wheel and the belt; 2. Check the cloth bag of the cloth bag dust collector (4301-DUC-201) and the ash in the horizontal flue in front of the cloth bag dust collector. 1. 检查胶带机(4303-BCV-002)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在； 2. 检查布袋除尘器 (4301-DUC-201)的布袋沾粘物料，布袋除尘器前水平烟道中的积灰。
7	Start-Up Of Rematerial Post 返料岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from back to front that is the reverse drive, each equipment needs to be switched on later; At the same time, confirm that the process behind the return post has been started and has the feeding conditions; Open the adhesive tape machine (4303-BCV-002) ⟹ Open the outlet valve of the fan (4301-FAN-201) (****), Open the fan (4301-FAN-201), when the fan operates normally without load, open the inlet valve of the fan (4301-FAN-201) (****) ⟹ Open the cloth bag filter (4301-DUC-201) ⟹ Open the star-type discharge valve of the cloth bag dust collector ⟹ Start the spiral conveyor in the cloth bag dust collector ⟹ Adjust the spray pressure of the bag filter, the setting point, pulse frequency and pulse width (after the first startup setting, the next startup will not adjust), start the bag filter ⟹ Open the inlet valve of the front end of the dust collector (******), and enter the dust and smoke gas ⟹ Observe the rising resistance of the dust collector and the discharge of the exhaust outlet. When the new filter bag is put into use, there will be trace dust escape from the

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№	Stage	Key Points
		discharge outlet⟹ When the resistance of the bag filter rises to 1000Pa~1200Pa, start the ash removal controller for ash removal, and observe the pressure drop. 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，各设备都需要延时开启；同时，确认返料岗位后面的工序已经开车并具备给料条件： 开启胶带机(4303-BCV-002)⟹打开风机 (4301-FAN-201)出口阀门(****)，开启风机 (4301-FAN-201)，待风机无负荷运转正常，打开风机 (4301-FAN-201)入口阀门(****)⟹开启布袋除尘器 (4301-DUC-201)开启布袋除尘器的耐温卸料器⟹启动布袋除尘器内的螺旋输送机⟹调整布袋除尘器喷吹压力、开始喷吹的设置点、脉冲频率和脉冲宽度(首次开机设定后，下次开机不再调整)，启动布袋除尘器⟹打开除尘器前端进口阀门(****)，通入含尘烟气⟹观察除尘器运行阻力上升情况及排气口的排放情况，在新滤袋投入使用时，排放口将会有微量粉尘逸出⟹当布袋除尘器阻力上升到1000Pa ~ 1200Pa时，启动清灰控制器进行清灰，并观察压力下降情况。
8	The Rematerial Post Stops 返料岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order and material discharge order, Each equipment needs to stop; Close the inlet valve of the fan (4301-FAN-201) (****) and stop the fan (4301-FAN-201), Close the outlet valve (****) of the fan (4301-FAN-201)⟹ Stop the bag dust collector before the bag filter (4301-DUC-201) Close the front inlet valve of the dust collector (****) and stop the dust and smoke gas⟹ After the collected dust and dust, Stop the spiral conveyor in the cloth bag dust collector⟹Stop the star-type discharge valve of the cloth bag dust collector⟹Stop the tape machine (4303-BCV-002)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序和物料排净顺序停车，各设备

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№	Stage	Key Points
		都需要延时停止： 关闭风机 (4301-FAN-201)入口阀门(****)，停止风机 (4301-FAN-201)，关闭风机 (4301-FAN-201)出口阀门(****)⟹布袋除尘器前端无含尘烟气后，停止布袋除尘器 (4301-DUC-201)关闭除尘器前端进口阀门(****)，停止含尘烟气⟹收集的烟尘清灰结束且输送完成后，停止布袋除尘器内的螺旋输送机⟹停止布袋除尘器的耐温卸料器⟹停止胶带机(4303-BCV-002)⟹打扫现场卫生。 2. Emergency Stop a. In an emergency, Stop the adhesive tape machine (4303-BCV-002), cloth bag filter (4301-DUC-2011), Fan (4301-FAN-201), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close the bag filter (4301-DUC-201) Front end inlet valve (****), Fan (4301-FAN-201) out and inlet valve (****,****), meanwhile, the upper level equipment shall stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, and the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，停止胶带机(4303-BCV-002)、布袋除尘器 (4301-DUC-201)、风机 (4301-FAN-201)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭布袋除尘器 (4301-DUC-201)前端进口阀门(****)、风机 (4301-FAN-201)出入口阀门(****,****) 同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生

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№	Stage	Key Points
		产，按照正常停车顺序进行操作。
9	Permits & isolation sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Rematerial Post 返料岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Rematerial posts in the chloride plant
CP 返料岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
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2					
3					

Approvals 批准

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